

ARTIFICIAL INTELLIGENCE

Fundamentals Of Artificial Intelligence

1.What is artificial Intelligence?

1. AI is a system or an algorithm that allows computers to perform tasks without explicitly being programmed to do so.
2. Artificial intelligence is the simulation of human intelligence processes by machines, especially computer systems.
3. AI which stands for artificial intelligence refers to systems or machines that mimic human intelligence to perform tasks and can iteratively improve themselves based on the information they collect.

Artificial intelligence (AI) makes it possible for machines to learn from experience, adjust to new inputs and perform human-like tasks.

Examples of AI-Artificial Intelligence

The following are the examples of AI-Artificial Intelligence:

1. Google Maps and Ride-Hailing Applications
2. Face Detection and recognition
3. Text Editors and Autocorrect
4. Chatbots
5. E-Payments
6. Search and Recommendation algorithms
7. Digital Assistant
8. Social media
9. Healthcare
10. Gaming
11. Online Ads-Network
12. Smart Home devices
13. Security and Surveillance
14. Smart Keyboard App
15. Smart Speaker
16. Smart Email Apps
17. Space Exploration

2. How AI works?

AI systems work by combining large sets of data with intelligent, iterative processing algorithms to learn from patterns and features in the data that they analyse. Each time an AI system runs a round of data processing, it tests and measures its own performance and develops additional expertise.

3. Purpose of AI

There are some main purposes and features in which artificial intelligence is used in different fields. These are the main purposes of artificial intelligence.

1. Improves decision making

The basic goal of artificial intelligence is to provide mechanism for decision making. This decision making is based on rare data as input data and will provide artificial intelligent result like human mind.

Artificial intelligence has the ability to make better decisions by automating the different physical and other tasks. These tasks can reduce the human labour and also saves the time.

2. Singularity

The ultimate objective of artificial intelligence is to overtake the work of human being. In near future, the growth of technology will become uncontrollable that will result into massive changes in human life style. Besides their side effects, these intelligent technologies will make the work simpler and efficient. It is also called technological singularity.

3. Machine learning

The main difference between machine learning and artificial intelligence is that machine learning is mainly concerned with accuracy. Machine learning is the sub-field of artificial intelligence and it takes data to produce the output as it is more focused.

4. Business process optimization

Business has the vital integrity in the economy of any country. The business process optimization is carried out by streamlining the work and removing the redundancies that ultimately results in improvement of the business.

The robotic process optimization is also used to minimize the daily routine work performed by the humans through different algorithms.

5.Creative work in technologies

There are number of technologies that are used to simplify the workflow and are easy to integrate across the business. These technologies are very important and are playing an important role in different fields of life. Some examples are:

1. Virtual reality
2. Live streaming apps.
3. Predictive Analytics.
4. Drones.
5. Motion animation.

6.Provide financial services

Artificial intelligence has played a huge role in financial services. It is used in fraud detection, risk management, asset management and insurance besides countless other sub-fields of financial services. There is almost no sub-field left without the use of artificial intelligence applications.

Similar to other fields the AI applications have saved a lot of time, human resources and made the results efficient.

7. Health care

It the most important sector where the artificial intelligence has revolutionized the sector. A large number of healthcare institutions are using the artificial intelligence machines for better and fast diagnoses the diseases in the patients.

IBM Watson is the one of the known and best healthcare technology which is based on simple questionnaire and responds according to the disease like x-ray reading.

Virtual assistant devices, based on artificial intelligence, are widely used in healthcare sector.

8. Automotive

Artificial intelligence has a huge impact on automotive industry. You will find it everywhere from car manufacturing industry to driver monitoring and driver recognition. There is artificial intelligence software available for driver monitoring. The software can make seat adjustment, mirror adjustment and even temperature adjustment.

The scenario of automotive industry has complete changed since the indulgence of artificial intelligence in the industry.

9. HR & Recruitment

Artificial intelligence in HR & recruitment is to boost-up the speed and precision of decision making and make the selection more reliable and accurate.

If the recruiter is not using the artificial intelligence in recruitment process, then the business is losing a lot of resources such as time and money.

So, the automation of workflow for recruitment process and management of the staff artificial intelligence applications plays a huge role and saves the resources.

5. Types of Artificial Intelligence

1. Reactive machines

Reactive machines are basic AI types that do not store past experiences or memories for future actions. Such systems zero in on current scenarios and react to them based on the best possible action. Popular examples of reactive machines include IBM's Deep Blue system and Google's AlphaGo.

2. Limited memory machines

Limited memory machines can store and use past experiences or data for a short period of time. For example, a self-driving car can store the speeds of vehicles in its vicinity, their respective distances, speed limits, and other relevant information for it to navigate through the traffic.

3. Theory of mind

Theory of mind refers to the type of AI that can understand human emotions and beliefs and socially interact like humans. This AI type has not yet been developed but is in contention for the future.

4. Self-aware AI

Self-aware AI deals with super-intelligent machines with their consciousness, sentiments, emotions, and beliefs. Such systems are expected to be smarter than a human mind and may outperform us in assigned tasks. Self-aware AI is still a distant reality, but efforts are being made in this direction.

6.Goals of AI

1. Develop problem-solving ability. ...
2. Incorporate knowledge representation. ...
3. Facilitate planning. ...
4. Allow continuous learning. ...
5. Encourage social Intelligence. ...
6. Promote creativity. ...
7. Achieve general intelligence. ...
8. Promote synergy between humans and AI.

7.Applications of AI

Artificial Intelligence (AI) is a branch of computer science that aims to create intelligent machines capable of performing tasks that typically require human intelligence. AI systems can learn from data, adapt to new information, and make decisions or solve problems.

Here are 10 applications of AI in various fields:

1.Healthcare:

- Medical Diagnosis: AI algorithms can analyze medical images, such as X-rays and MRIs, to assist doctors in diagnosing diseases like cancer.
- Drug Discovery: AI accelerates drug discovery by predicting potential drug candidates and their efficacy.

2. Finance:

- Algorithmic Trading: AI-driven algorithms analyze financial data and execute trades at high speeds, optimizing investment portfolios.
- Fraud Detection: AI can identify fraudulent transactions by detecting unusual patterns and behaviours in financial data.

3. Autonomous Vehicles:

- Self-Driving Cars: AI enables cars to perceive their environment, make real-time decisions, and navigate autonomously.

4.Natural Language Processing (NLP):

- Chatbots: AI-powered chatbots engage in natural language conversations with users to provide customer support or answer questions.
- Language Translation: NLP models like Google Translate can translate text and speech between languages.

5. Education:

- Personalized Learning: AI systems adapt educational content and assessments to individual student needs.
- Automated Grading: AI can automatically grade assignments and provide instant feedback to students.

6.E-commerce:

- Recommendation Systems: AI algorithms analyze user behavior and preferences to recommend products and content.
- Inventory Management: AI optimizes inventory levels and supply chain logistics.

7. Manufacturing:

- Quality Control: AI-powered robots and cameras inspect products for defects in real-time.
- Predictive Maintenance: AI predicts equipment failures and schedules maintenance to minimize downtime.

8. Robotics:

- Industrial Robots: AI-driven robots perform tasks like welding, assembly, and pick-and-place operations in manufacturing.
- Social Robots: AI-powered robots interact with humans in settings like healthcare, customer service, and companionship.

9. Agriculture:

- Precision Agriculture: AI analyzes data from sensors, drones, and satellites to optimize crop management, irrigation, and yield prediction.

10. Entertainment:

- Content Creation: AI can generate music, art, and even write articles or scripts.
- Gaming: AI enhances video game experiences through intelligent opponents and dynamic storytelling.

11. Data analysis and pattern recognition:

AI excels in analysing complex dataset and identifying the pattern that humans may not easily detect. This capability is usually valuable in the field of economics, finance healthcare marketing where data-driven insights are crucial.

12. AI in Travel & Transport

AI is becoming highly demanding for travel industries. AI can do various travel related works such as from making travel arrangement to suggesting the hotels, flights, and best routes to the customers. Travel industries are using AI-powered chatbots which can make human-like interaction with customers for better and fast response.

8.Ethics in AI

- AI ethics is a system of moral principles and techniques intended to inform the development and responsible use of artificial intelligence technology. As AI has become integral to products and services, organizations are starting to develop AI codes of ethics.
- An AI code of ethics, also called an AI value platform, is a policy statement that formally defines the role of artificial intelligence.
- Some of the AI ethics are:
 1. **Explainable:** AI system should be transparent, particularly about what goes into its algorithm's recommendations.
 2. **Robustness:** AI-powered systems must be actively defended from adversarial attacks,
 3. AI system should minimize security risks and enabling confidence in system outcomes.
 4. **Transparency:** To reinforce trust users must be able to see how the service works, evaluate its functionality, and comprehend its strengths and limitations.
 5. **Privacy:** AI systems must prioritize and safeguard consumers' privacy and data rights.

9.Significance of data in AI

Data is important factor of AI. The quality and depth of data will determine the level of AI applications the system can achieve. Hence it is required the data to meet following characteristics:

- **It is complete:** There Should not be empty spots or cells in datasets. Every slot should have a piece of data in it, and there are no visible holes in them.
- **It is comprehensive:** The datasets are as complete as they can be.
- **It is consistent:** All the datasets must fit under the variables that it is has been assigned to.
- **It is accurate:** The data should be collected from the proper source and it should be accurate because data is selected for various feeds for our AI system, the data source must be trusted. If there are chunks that are not accurate, the output will be skewed, and you will not get a correct answer.
- **It must be valid:** This is crucial with time series datasets. There may be old data that could interfere with the learning process of the AI system when analysing recent datasets. The system should always learn from the recent data.
- **It is unique:** Each piece of data must be unique to the variables it is serving.

10.AI Software Development life cycle

1. The AI life cycle involves various stages from data collection, data analysis, feature engineering and algorithm selection to model building, tuning, testing, deployment, management, monitoring and feedback loops for continuous improvement.
2. Whereas the software development tools and processes are standardized with DevOps.
3. Agile methodologies are used during the AI developmental life cycle.
4. However, with the breadth and depth of AI disciplines, frameworks and languages, specialized capability is required and specialized tools need to come in, to manage every stage of AI project development.
5. The software space is also fragmented, with tools from large companies to small start-up companies.
6. Based on our interactions with clients, the enterprise has seeking adopting of end-to-end AI life cycle development tools including H2O.ai, Kubeflow, and MLflow. however, there is a long way to go, as standardization of these tools and pipelines is still a work in progress.

11.Compare between AI software developmental life cycle and traditional software developmental life cycle.

AI Software Development Life Cycle	Traditional Software Development
• Data-Driven AI software development is heavily dependent on data and requires large amounts of high-quality data to train and test models	• Data-Driven In traditional software development, data is often an afterthought.
• Experimentation and Iteration AI software development often involves a lot of experimentation and iteration, as different algorithms and approaches are tried and tested to see which ones work best	• Experimentation and Iteration Traditional software development is typically more linear and follows a specific plan or design.
• Model Selection In AI software development, selecting the right model for a particular problem is critical and can be a time-consuming process.	• Model Selection In traditional software development, the choice of algorithms and techniques is often predetermined

Model evaluation and performance In AI software development, model performance is evaluated using different metrics and techniques, such as accuracy, precision, recall, and F1 score	Model evaluation and performance In traditional software development, model evaluation is often based on functional requirements.
Deployment and Maintenance AI software deployment and maintenance requires additional considerations, such as retraining models over time and deploying them in production environments	Deployment and Maintenance In traditional software development, deployment and maintenance are often simpler and more straightforward.
But in Machine Learning, a model is built from the data, and that model is the logic	In traditional programs, a developer designs logic or algorithms to solve a problem. The program applies this logic to input and computes the output.
Explainability: AI models are often complex and difficult to understand, which can make it challenging to explain their predictions and decision to non-technical stake holders	In traditional software development explainability is not as much of an issue.

AI influence in companies viz, Amazon, Microsoft, Google, IBM

Google:

- AI is embedded in Google's core. Google's search engines, Google Maps, Google Photos, to YouTube and Gmail Smart Reply, AI is integrated inside all of the Google apps we use on a daily basis.
- Google is driving AI through its technological offerings like TensorFlow and Cloud AutoML (to name a few) that is helping developers to innovate by using its open-source projects.
- Google Brain, the AI-based technology at the foundation of YouTube monitors and records users' viewing habits as they use the app to stream content.
- Google's Cloud Video Intelligence segments and analyses all the videos stored on Google for context and content to generate video summaries automatically and alerts when the system notices suspicious activities.

Amazon

- It is one of the first companies to provide collaborative filtering-based personalized recommendations of the products.
- Amazon's "Anticipatory shipping" that allows products to reach our nearest possible location before we place the order to purchase the product. It calculates the number of drivers required to make deliveries using an app called Flex that depends on AI. The app considers many factors like the number of packages in the same locality, the weight of the package, etc to make every minute count.
- Amazon also uses an Ai-powered dynamic pricing algorithm to get an edge over competitors. The algorithm uses AI to enable optimal sales and revenue automatically by decreasing the prices of the products to increase sales when it's needed and vice-versa.
- Amazon's AI-backed recommendation engines algorithm uses data from customer's previous purchases, preferences, browsing history, search history to create a list of personalized products that a customer will likely buy.

Microsoft

- Microsoft has launched many AI solutions across industries like healthcare, retail, education, banking, etc. One of the oldest contenders, here is Microsoft's AI products and plans.
- Microsoft uses AI to fight against cybercriminals by learning from the data of every company that uses its services.
- The company's Azure security team customizes security to a client's online behaviour.
- Microsoft is bringing a change to the healthcare industry by helping to find the best cancer treatment. A group of researchers have created algorithms to understand how cancer develops and predict the best drug combinations to fight it.
- Carlsberg, a leading beer and beverages company, uses Microsoft's AI solutions to detect various aromas and flavours of beer, and also improve the quality of existing beers.

IBM

- The ThreeDWorld Transport Challenge.

This AI innovation project is tasked with improving research on AI agents that can control a simulated mobile robot guided by computer vision to manipulate objects, pick them up, and carry them to new locations. The agent must locate small objects distributed around the simulated house, pick them up, and take them to a new, final spot.

- Neuro-Symbolic Artificial Intelligence.

This AI innovation involves combining neural networks that pull statistical structures from raw data (e.g., image files), with symbolic representations of logic and problems. Ultimately, researchers hope

to make machines learn more like people do—mastering abstract concepts and connecting words with images. Researchers are working on giving machines actual common sense and reasoning capabilities.

- **The Watson Factor**

Watson is IBM's question-and-answer computer system, made up of a portfolio of business-ready applications, solutions, and tools.

Watson was created to reduce costs and AI obstacles while optimizing the responsible use of AI and its outcomes. Watson helps organizations automate complex processes, predict future outcomes, and optimize employees' time.

The evolution of AI

- **1950: The Turing Test** – The Turing test marked the inception of AI, when Alan Turing conducted a test to examine the machine's ability to exhibit intelligent behaviour that is similar to, to indistinguishable from a human.
- **1956: Birth of the term 'AI'**- The term AI was first coined at the Dartmouth Conference by American computer scientist John McCarthy. This was the first time AI was considered as an academic field. Also, the same year, Allen Newell, Herbert A. Simon, and Cliff Shaw invented 'Logic Theorist'- the world's first AI program.
- **1961: Introducing Unimate**– The year marked the emergence of Unimate, the world's first industrial robot that worked for General Motors, and undertook the job of transporting die castings from an assembly line and welding them on auto bodies, a dangerous job for humans.
- **1972: WABOT 1**- The first intelligent humanoid robot was developed in Japan, consisting of a limb-control system, a vision system, and a conversation system. It was the first fun-scale anthropomorphic robot.
- **1997: Deep Blue**– Deep Blue was a chess-playing expert system, run on a unique purpose-built IBM supercomputer. It was the first computer to win a match against a world champion with regular time controls.
- **2002: Roomba**: World's first mass-produced autonomous vacuum cleaner, from iRobot Corporation that could navigate and clean homes on its own. They possess sensors that enable them to detect obstacles and dirt spots on the floor.

- **2010: WATSON– IBM** launched its first supercomputer that combines artificial intelligence and other technologies, enabling organizations to predict future outcomes, automate complex processes, and optimize employees' time.
- **2011: Siri**– Apple launched its revolutionary chatbot and virtual assistant that is a part of all Apple electronic products. It can support a wide range of user commands and was the first of its kind back in 2011.
- **2016: Sophia**- Hanson Robotics created a humanoid robot named Sophia who is also known as the first 'robot citizen'. Sophia's similarity to an actual human being is what sets her apart from previous humanoid robots.
- **2018-2020**- AI inventions in Gastroenterology– Within this time period, there have been several innovations in the AI domain related to healthcare services, especially in gastroenterology to enhance accuracy in treatments.
- **2021: Ethical AI**- Ethical AI became an important topic in 2021 for business leaders who were looking to get the most out of AI solutions and keep the customer happy simultaneously.

How AI will Impact on daily life, work life, work force, jobs, products and services??

How will AI affect your everyday life?

1. Artificial intelligence has made a huge impact on the way we live our lives. From AI-powered cars to voice assistants like Alexa and Siri, that help you with everything from managing your schedule to finding a good restaurant nearby.
2. AI will play a more pivotal role in the medical field, with machines making diagnoses and suggesting treatments. Soon, this technology could be used to help diagnose diseases like Parkinson's or Alzheimer's.
3. As AI becomes more integrated into our lives, it will be used to automate things like managing personal finances and filing taxes
4. As AI starts to play a bigger role in our day to day lives, it will also be used for more sophisticated tasks like driving and construction.
5. Your car will drive you around town instead of you doing so yourself
6. You'll never have to go grocery shopping again because your fridge will automatically order groceries for when they're running out

AI Impact on Jobs

1. Blue-collar professions and customer service jobs are considered to be amongst the first that will be replaced with robots and A.I.
2. The main advantage that A.I. has over humans is the incredible amount of processing power and access to a vast storage of raw data. With the right input and guidance input, A.I. can make much more accurate analysis and predictions than a human expert, at a fraction of a time.
3. This means that even white-collar workers, such as financial analysts, could lose their jobs, or parts of them, to A.I. Even the doctors are potentially in jeopardy because A.I. can utilize the patient's data in such a way as to potentially prescribe the ideal medicine and therapy for each patient.
4. However, A.I. at this stage still relies heavily on human guidance. That is why we can expect that in the next 10 years, A.I. will become more of an invaluable assistance than a threat to most jobs.

AI Impact on New services and products

1. The speed with which media, telecommunications, and technology industries can offer new products and services is expected to increase by 52%.
2. The Marketing industry can also look forward to an exponential development of A.I.
3. For example, Facebook and YouTube are offering ads or songs that cater to our tastes with increasing accuracy.
4. As A.I. advances, more accurate and personalized targeting will take place, which will, in turn, improve both customer experience and product sales.

The evolution of HMI

- The evolution of the Human Machine Interface, HMI, had a rapid acceleration in recent. Touch-screen technology is the most widespread HMI application, but since 2018 there have been developed new important technologies like conversational interface and AR and VR technologies.
- The rapid development of HMIs has influenced interface design in the consumer and industrial areas.
- To guarantee a smooth and efficient relationship between humans and machines, it is fundamental to use high-quality hardware, like boards and processors, that can support the optimal functionality of the device.
- The clients, whether they are smartphone users or workers on an industrial machine, want a service or a product that is fast, intuitive, and easy, if not natural, to use.

HMI in today's industry

- In today's industry, HMIs are mainly touch-screen and can be used in different situations. The industrial areas that have integrated innovative Human Machine Interfaces are the automotive, pharma, transport, and energy sector.
- HMI focus needs to be usability, flexibility, speed, and security. A Human Machine Interface must be easy to use, adaptable if needed, fast, and secure to use.

Human Machine Interface: from touch-screen to “Mind and Body”

- The HMI technological evolution will not stop with touch screens. From the “Hands & Touch” era, when the command was given by pushing a button or as recently with a simple touch, we will move to the “Mind & Body” era, where our body is the interface and our mind will guide the actions. This shift will bring the need for more immersive and smart interfaces.

Important questions (10 Marks)

1. What is AI? what are the different types of AI? Explain the applications of AI?
2. With Industry 4.0, artificial intelligence is finding place in every aspect of life. What happens if AI replaces humans in the workplace?
3. How AI will Impact on daily life, work life, work force, jobs, products, and services?
4. In recent years most of the companies are turning to text based chatbot for resolving consumer queries. What is the reason for it and how is it impacting the business. (5 Marks)

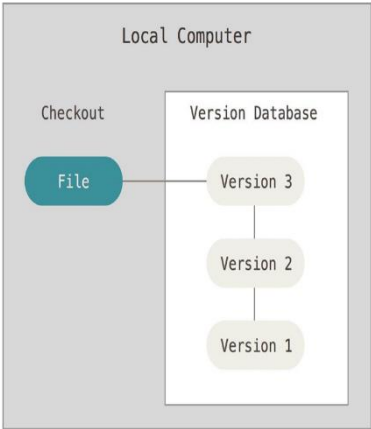
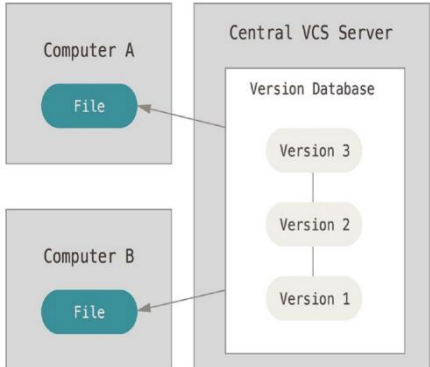
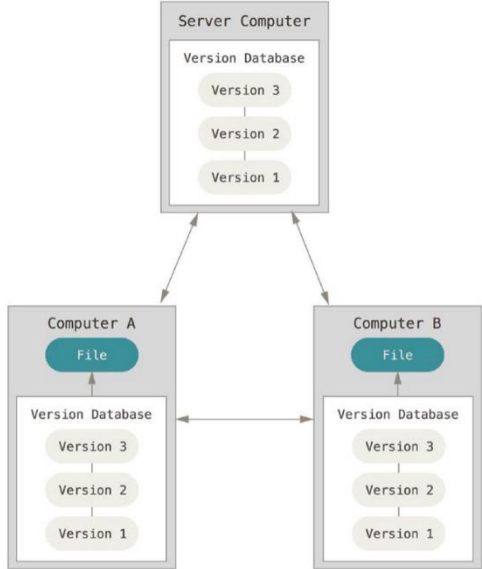
Exams questions:

1. Artificial intelligence (AI) is a promising state-of-art technology that provides intelligent solutions in every field today. Justify your answer by describing AI and its application in various fields.
2. Write steps to create repository in GitHub and add file.
3. How is AI software development different from traditional software development? Explain.

Fundamentals of GIT & GIT HUB

Why Do We Need a Version Control System?

- Version control is a system that records changes to a file or set of files over time so that one can recall specific versions later.
- In software development, versioning **allows development teams to keep track of changes they make to the project code**. The changes may include new functions, features, or bug fixes. Minor changes can also be tracked in a similar manner.

1. <u>Local Version Control Systems</u>	2. <u>Centralized Version Control Systems</u>	3. <u>Distributed Version Control Systems</u>
In this version-control method the choice is to copy files into another directory. This approach is very common because it is so simple, but it is also error prone. It is easy to forget which directory you're in and accidentally write to the wrong file or copy over files you don't mean to. The next major issue that people encounter is that they need to collaborate with developers on other systems.  Figure 1-1. Local version control	The next major issue that people encounter is that they need to collaborate with developers on other systems. To deal with this problem, Centralized Version Control Systems (CVCSs) were developed. These systems, such as CVS, Subversion, and Perforce, have a single server that contains all the versioned files, and a number of clients that check out files from that central place. This setup also has some serious downsides. The most obvious is the single point of failure that the centralized server represents.  Figure 1-2. Centralized version control	In a DVCS (such as Git, Mercurial, Bazaar, or Darcs), clients don't just check out the latest snapshot of the files—they fully mirror the repository. Thus, if any server dies, and these systems were collaborating via it, any of the client repositories can be copied back up to the server to restore it. Every checkout is really a full backup of all the data. 

What is Git?

Git is an open-source distributed version control system. It is designed to handle minor to major projects with high speed and efficiency. It is developed to co-ordinate the work among the developers. The version control allows us to track and work together with our team members at the same workspace.

Installing Git

- There are also a few ways to install Git on Windows. The most official build is available for download on the Git website. <http://git-scm.com/download/win> and the download will start automatically.
- Note that this is a project called Git for Windows (also called msysGit), which is separate from Git itself; for more information on it, go to <http://msysgit.github.io/>.

Your Identity

- The first thing after installing Git is to set user name and e-mail address.
- This is important because every Git commit uses this information, and it's immutably baked into the commits that are created.

To tell git about user

- `$ git config --global user.name "shilpaspt"`
- `$ git config --global user.email "shilpasiet@gmail.com"`
- `$ git config --global --list` # To check the info you just provided

Getting a Git Repository

Initializing a Repository in an Existing Directory

```
$ git init          # initialises an empty git repository
```

The above command will create an empty. git repository. Suppose we want to make a git repository on our desktop. To do so, open Git Bash on the desktop and run the above command.

Create a new directory to initialise the git repo

```
$ mkdir project      #makes new directory project
$ cd project
$ git init           # initialises the git repository
$ vi f1.txt          # To create a f1.txt file in the repository
```


Add files to the Staging Area for commit:

```
$ git log           #shows the status of the f1.txt file as untracked to stag the changes
$ git add .         # Adds all the files in the local repository and stages them for commit
$ git add f1.txt     # To add a specific file
```

Before committing, to see what are all the files that staged

```
$ git status        # Lists all new or modified files to be committed
```

To Commit the changes made to git repo

```
$ git commit -m "First commit"
```

The message in the " " is given so that the other users can read the message and see what changes is made

Add a remote origin and Push:

```
$ git remote add origin remote_repository_URL    # sets the new remote
```

The **git remote** command lets us create, view, and delete connections to other repositories.
The **git remote -v** command lists the URLs of the remote connections we have to other repositories.

```
$ git remote -v      # List the remote connections you have to other repositories.
```

Now the **git push** command pushes the changes in the local repository up to the remote repository that is specified as the origin.

```
$ git push -u origin master    # pushes changes to origin
```

Make the changes in the git repo (changes made to the file f1.txt second time)

```
$ echo I have changed the file >file1.txt
$ git add .
$ git commit -m "second change"
$ git push -u origin master
```

viewing the history of all the changes

```
$ git log           # to view the history of all the changes made on to the file.
```

Cloning a Git Repo:

\$ git clone # clones the remote repository to the current working directory. The changes to the file can be made and updated using pull command.

Git Branching and merging

Branch allows to work on different versions of the same file in parallel. The changes made in one branch can be independent of the other. later we can decide to merge changes to other branches. Branches results in the separation of version of same files.

By default, the repository is in the master branch and Head is a pointer that normally pointing to a branch.

\$ Git branch # return the current working branch information

Creating and switching to new branches

\$ git branch branchname # creates new branch
\$ git branch SDN # SDN branch is created
\$ Git branch Auth # auth branch is created
\$ git branch # return all the branches
 # Git branch head pointing to master branch

Switching between branches

\$ git checkout SDN # switched to the branch SDN from master
\$ git status
\$ vi f1.txt # alter the f1.txt created in master branch
\$ git add si.txt # stag and commit the changes made to the file f1.txt
\$ git commit -m "sdn change"
\$ git status

Switching to auth branch

\$ git checkout auth # switched to the branch auth from master
\$ git branch
\$ vi f1.txt # alter the file f1.txt in auth branch
\$ git status # shows the file f1.txt as modified and not committed
\$ git add s1.txt
\$ git commit -m "change in auth" #stag and commit the changes made to file f1.txt in auth
 branch

Merging local branches together

Fast forward merge:

\$ git merge branchname
\$ git merge SDN # merges the master branch with SDN branch

\$ type f1.txt

merged file f1.txt contains the changes made in SDN branch for the same file

\$git merge auth

#merges the auth branch with master. (it is a kind of force merge)

Deleting branches

\$ git branch --merged

returns the details of the branch that are merged

\$ git branch -d SDN

deletes the branch SDN

\$ git branch -d auth

deletes the branch auth

\$ git checkout -b dev

GitHub

GitHub is an online software development platform used for storing, tracking, and collaborating on software projects. It enables developers to upload their own code files and to collaborate with fellow developers on open-source projects. GitHub also serves as a social networking site in which developers can openly network, collaborate, and pitch their work.

Account creation and configuration

- *Create a new account in the github using valid email-id*
- *To create new repository*

The screenshot shows the GitHub 'Start writing code' interface. On the left, under 'Start a new repository', there is a text input field for the repository name, currently showing 'shilpaspt /'. Below this, there are two radio button options: 'Public' (with a lock icon) and 'Private' (with an open lock icon). The 'Private' option is selected. A green 'Create a new repository' button is at the bottom. On the right, under 'Introduce yourself with a profile README', there is a text area for the README content, currently showing a list of five items. A green 'Create' button is at the bottom right of the README section.

- *Give a suitable repository name and check the readme file and click on the create repository*

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere? [Import a repository.](#)

Owner * / Repository name * ✓

Great repository names are short and memorable. Need inspiration? How about [improved-octo-funicular?](#)

Description (optional)

☒ **Public**
 Anyone on the internet can see this repository. You choose who can commit.

☐ **Private**
 You choose who can see and commit to this repository.

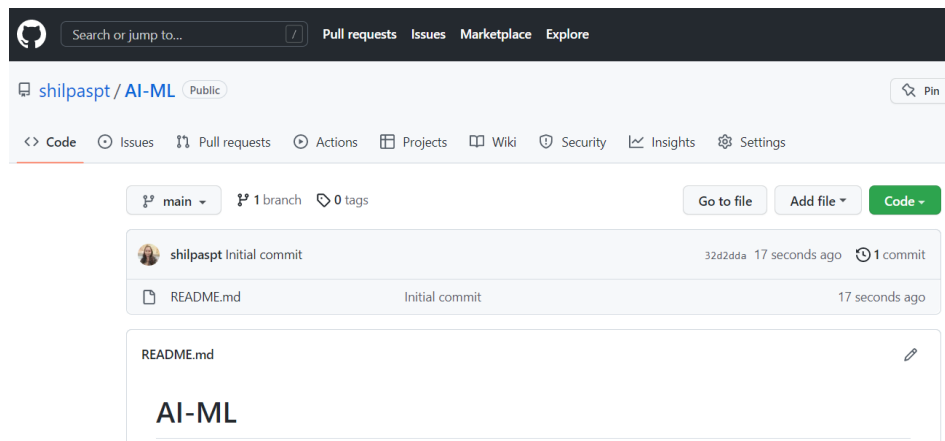
Initialize this repository with:
 Skip this step if you're importing an existing repository.

☒ **Add a README file**
 This is where you can write a long description for your project. [Learn more.](#)

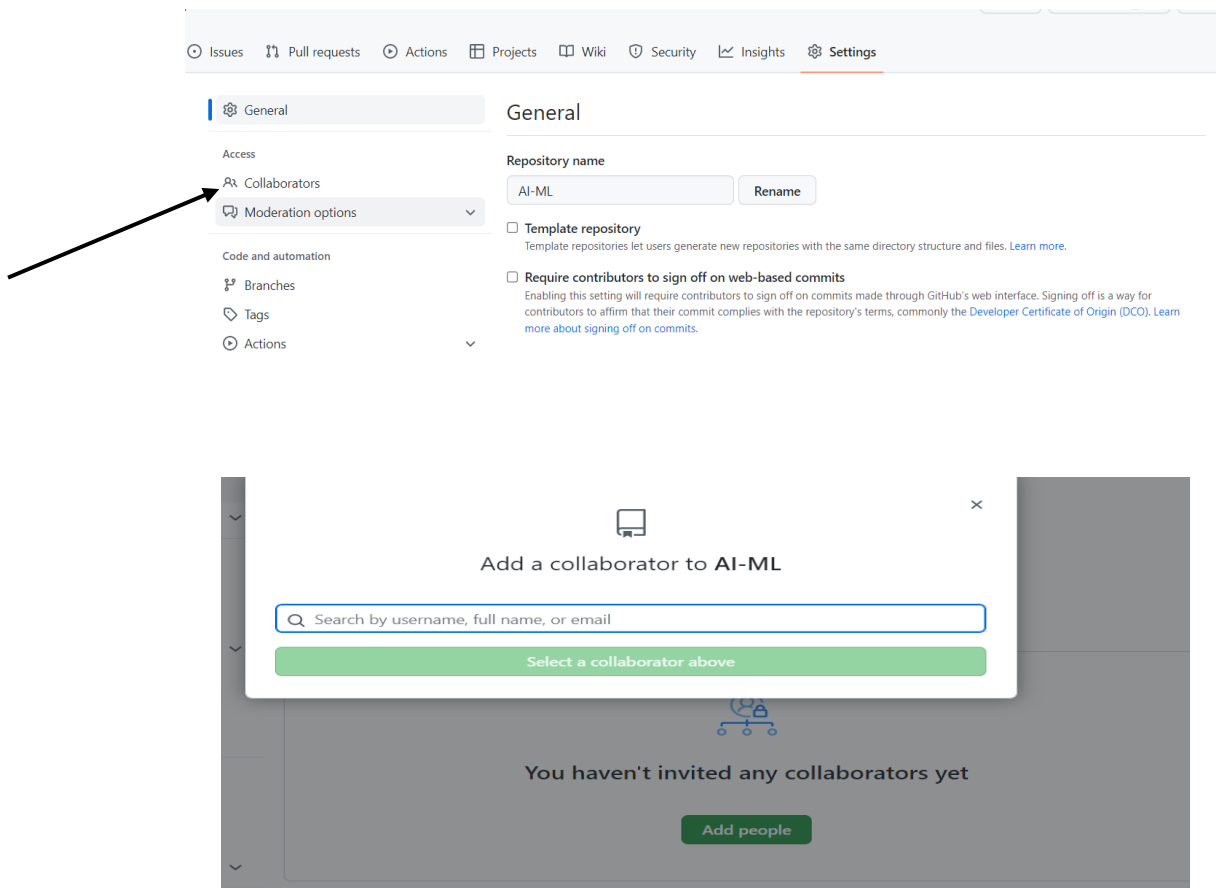
Add .gitignore
 Choose which files not to track from a list of templates. [Learn more.](#)

.gitignore template: **None** ▼

- *New repository AI-ML has been created*



- *Repositories can be clone by using the fork option, it allows developers to clone others entire repository to their github account.*
- *Pull request enables the user to request for the changes made the repository to be accpeted by the repository owners.*
- *Collabroration allows the users/developers to add collabrtrate developers ,so that they can make any changes to the file or code and file can be accessed simultaneously.*



- *GitHub Collaboration allows different people can work on the same project together. It is like creating a group in GitHub just like Groups in other social media. The people added to the collaborator's list can be able to push, merge, and do other kinds of similar things on the project.*
- *The Migration API allows to migrate the repositories and users of one organization from GitHub.com to GitHub Enterprise Server.*